THE DIAGONAL OF AN ALGEBRAIC SPACE IS LOCALLY OF FINITE TYPE, A MONOMORPHISM, SEPARATED AND LOCALLY QUASI-FINITE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains one xymatrix display; it is part of the counted proof span and is preserved verbatim, with the chapter's own Xy-pic setup retained in the preamble.

Editorial note. The seven references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- The cartesian square computing the diagonal from a presentation. used at proof lines 2311-2318. Reduces the properties of Delta to those of j.
- Properties of representable morphisms of spaces. Lemma lemma-representable-morphisms-spaces-property (used at proof lines 2319-2321), together with Remarks remark-list-properties-stable-base-change and remark-list-properties-fpqc-local-base. Transfers the four properties between Delta and j.
- A monomorphism of schemes is separated. Schemes, Lemma schemes-lemma-monomorphism-separated, used at proof lines 2325-2327. Gives separatedness of j.
- Permanence of finite type and finite fibres. Morphisms, Lemmas morphisms-lemma-permanence-finite-type and morphisms-lemma-finite-fibre, used at proof lines 2328-2334. Give local finiteness of type and local quasi-finiteness of j.

Source reference. Original lemma 0265: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2838–2838. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3822–3822. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3479–3479. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01T8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2365–2365. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 02WH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3804–3804. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original remark 02WF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3802–3802. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01L4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2073–2073. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let F be an algebraic space over S. Let $\Delta : F \to F \times F$ be the diagonal morphism. Then*

(1) *Δ is locally of finite type,*
(2) *Δ is a monomorphism,*
(3) *Δ is separated, and*
(4) *Δ is locally quasi-finite.*

Proof. Let $F = U/R$ be a presentation of F. As in the proof of Lemma 0265 the diagram

$$\begin{array}{ccc} R & \longrightarrow & F \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle \Delta} \\ U \times_S U & \longrightarrow & F \times F \end{array}$$

is cartesian. Hence according to Lemma 02WZ it suffices to show that j has the properties listed in the lemma. (Note that each of the properties (1) – (4) occur in the lists of Remarks 02WF and 02WH.) Since j is an equivalence relation it is a monomorphism. Hence it is separated by Schemes, Lemma 01L4. As R is an étale equivalence relation we see that $s, t : R \to U$ are étale. Hence s, t are locally of finite type. Then it follows from Morphisms, Lemma 01T8 that j is locally of finite type. Finally, as it is a monomorphism its fibres are finite. Thus we conclude that it is locally quasi-finite by Morphisms, Lemma 02NG. □